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INTRODUCTION TO PASSIVE DECAY HEAT REMOVAL SYSTEMS

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Introduction

Passive safety systems represent a key phenomenon of modern nuclear reactor design, particularly in the context of Decay Heat Removal following reactor shutdown. After a reactor is scrammed, residual heat generated by radioactive decay continues to be produced within the fuel. The reliable removal of this decay heat is essential to prevent fuel overheating and to ensure the integrity of the reactor core and containment systems.

Among the various technological solutions adopted in advanced reactor concepts, Passive Decay Heat Removal (PDHR) systems exploit fundamental physical principles (such as gravity, density gradients, and natural circulation...) to guarantee core cooling without reliance on active components, external power supplies, or operator intervention. This approach significantly enhances plant safety by eliminating potential common cause failures associated with active safety systems.

Natural circulation is a thermo-hydraulic phenomenon driven by buoyancy forces that arise from density differences within a fluid subjected to temperature gradients. In closed loop configurations the density difference between a heated (hot) leg and a cooled (cold) leg generates a hydrostatic pressure imbalance that sustains fluid motion. The resulting flow rate is determined by the equilibrium between the driving buoyancy head and the total pressure losses occurring along the loop, including distributed friction losses and localized losses at singular components.

The present assignment aims to develop a simplified yet physically consistent model of a natural circulation loop operating at saturation conditions and constant pressure. The system is designed to remove a prescribed thermal power through phase change, with saturated liquid entering the heater and saturated vapor entering the cooler. The objectives include the determination of the required elevation difference to sustain the imposed thermal load, the investigation of geometric design constraints, and the assessment of surface roughness effects on system performance.

Through this assignment, the fundamental engineering principles governing passive thermal-hydraulic systems are explored, highlighting both the analytical formulation of the problem and the practical aspects related to numerical implementation and design verification. The study provides essential insight into the thermo-fluid dynamic constraints that guide the conceptual design of passive safety systems in nuclear power plants.

1 Problem Statement

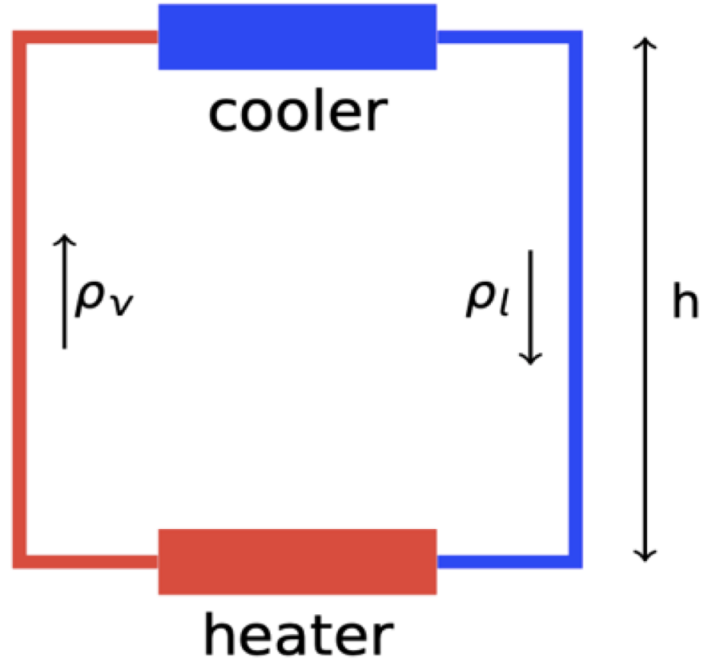


Figure 1: Scheme of a rectangular loop where the fluid flow is driven by buoyancy effects

Data:

- Thermal Power: $P_{th} = 34.8 \text{ MW}_{th}$
- Cooler pressure drop coefficient: $k = 20$
- Heater pressure drop coefficient: $k = 40$
- Pipe section length: $L = 10 \text{ m}$
- Pipe diameter: $D_e = 0.27 \text{ m} (10.750 \text{ in})^1$
- Pipe thickness: $t = 9.27 \text{ mm} (0.365 \text{ in})$

Assumptions:

- Both the heater and the cooler are assumed to be horizontal (i.e., constant elevation)
- Heat transfer in both the hot leg and cold leg is neglected
- System is in steady state
- Steam and liquid are always at saturation
- Constant pressure of $p_l = p_v = 70 \text{ bar}$ for liquid and steam phase

¹Pipe diameters were selected in accordance with the ANSI B36.10M standard. This approach ensures industrial consistency and provides standardized dimensions for welded and seamless wrought steel pipes, facilitating accurate hydraulic calculations and structural integration.

2 From the Navier-Stokes Equations

General motion of an incompressible Newtonian fluid is governed by the continuity equation (or conservation of mass) and the momentum equation:

$$\nabla \cdot \mathbf{V} = 0 \quad (1)$$

$$\frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V} \cdot \nabla) \mathbf{V} = -\frac{\nabla p}{\rho} + \mathbf{g} + \nu \nabla^2 \mathbf{V} \quad (2)$$

For steady, fully developed flow in a pipe, the velocity contains only an axial component, which is a function of only the radial coordinate $[\mathbf{V} = u(r)\hat{\mathbf{i}}]$. For such conditions, the left-hand side of the Eq.2 is equal to zero. This is equivalent to saying that the fluid experiences no acceleration as it flows along. Thus, with $\mathbf{g} = -g\hat{\mathbf{k}}$, the Navier-Stokes equations become:

$$\nabla \cdot \mathbf{V} = 0$$

$$\nabla p + \rho g \hat{\mathbf{k}} = \mu \nabla^2 \mathbf{V} \quad (3)$$

From Eq.3 the flow is governed by a balance of pressure, weight, and viscous forces in the flow direction. If the flow were not fully developed it would not be possible to simplify the Navier-Stokes equations (the nonlinear term $(\mathbf{V} \cdot \nabla) \mathbf{V}$ would not be zero), and the solution would be very difficult to obtain. Because of the assumption that $\mathbf{V} = u(r)\hat{\mathbf{i}}$, the continuity equation, Eq 1, is automatically satisfied. This conservation of mass condition was also automatically satisfied by the incompressible flow assumption. [1]

To derive the governing equations for the problem, and considering the closed-loop nature of the circuit, the circulation is applied to each term of Eq. 3 along the four segments comprising the path Γ :

$$\oint_{\Gamma} \nabla p \cdot d\mathbf{l} + \oint_{\Gamma} \rho g \hat{\mathbf{k}} \cdot d\mathbf{l} = \oint_{\Gamma} (\mu \nabla^2 \mathbf{V}) \cdot d\mathbf{l}$$

To solve the governing equation, an analysis of its three constituent terms is conducted:

- **Pressure term:** Since pressure is a state function, the line integral of its gradient along a closed path Γ is identically zero:

$$\oint_{\Gamma} \nabla p \cdot d\mathbf{l} = 0$$

- **Buoyancy term ($\Delta p_{buoyancy}$):** The integral of the body force along the circuit is non-vanishing if the density ρ varies (e.g., due to temperature gradients between the hot and cold legs). This term represents the driving head of the flow:

$$\oint_{\Gamma} \rho g \hat{\mathbf{k}} \cdot d\mathbf{l} \rightarrow \Delta p_{buoyancy}$$

- **Viscous term (Δp_{loss}):** The integral of the viscous forces represents the total hydraulic resistance encountered by the fluid due to wall friction and dissipation:

$$\oint_{\Gamma} (\mu \nabla^2 \mathbf{V}) \cdot d\mathbf{l} \rightarrow -\Delta p_{loss}$$

Consequently, the momentum balance derived from the Navier-Stokes equations for natural circulation flow within a closed loop reduces to the following:

$$\Delta p_{buoyancy} = \Delta p_{loss} \quad (4)$$

To derive the final governing equation, the previously defined integrals must be evaluated by applying classical fluid-dynamics models. Starting with the force integral along the circuit, $\oint_{\Gamma} \rho g \hat{\mathbf{k}} \cdot d\mathbf{l}$, in a natural

circulation loop, that exploits the density differential between the cold leg (liquid phase) and the hot leg (vapor phase), the integral reduces to the hydrostatic pressure difference between the two fluid columns²:

$$\Delta p_{buoyancy} = \int_{down} \rho_l g dz - \int_{up} \rho_v g dz = (\rho_l - \rho_v)gh \quad (5)$$

The viscous term in the Navier-Stokes equations, $\mu \nabla^2 \mathbf{V}$, when integrated along the circuit, represents the energy dissipated due to friction. In fluid dynamics, this dissipation is categorized into two main types:

- **Distributed Losses** (Δp_{distr}): These occur along the straight sections of the piping due to continuous friction between the fluid and the internal pipe walls. By integrating the velocity profile (for both laminar and turbulent regimes), the Darcy-Weisbach equation is obtained:

$$\Delta p_{distr} = f \frac{L}{D} \frac{1}{2} \rho v^2 \quad (6)$$

- **Concentrated Losses** (Δp_{conc}): These occur at specific points in the circuit (such as bends, valves, or transitions), where the fluid undergoes abrupt changes in direction or velocity. This leads to the formation of vortices and localized energy dissipation, which are modeled as:

$$\Delta p_{conc} = k \frac{1}{2} \rho v^2 \quad (7)$$

Ultimately, the momentum balance for a closed-loop system under natural circulation conditions is established by equating the buoyancy driven pressure head with the total hydraulic losses. The resulting governing equation is expressed as follows:

$$(\rho_l - \rho_v)gh = \sum f \frac{L}{D} \frac{1}{2} \rho v^2 + \sum k \frac{1}{2} \rho v^2 \quad (8)$$

In the present case, this equation is expanded by separating the contributions associated with the liquid phase (cold leg) and the vapor phase (hot leg), consistently with the physical configuration of the system.

Distributed losses are evaluated independently in the two sections of the loop. According to the problem assumptions, the fluid flows as saturated liquid in the cold leg and as saturated vapor in the hot leg. Therefore, two distinct contributions must be considered:

- Liquid phase (cold leg): characterized by friction factor f_l , length L_l , density ρ_l , and velocity v_l ;
- Vapor phase (hot leg): characterized by friction factor f_v , length L_v , density ρ_v , and velocity v_v .

Localized losses are associated with the main components of the system: the cooler and the heater:

- Cooler: the loss coefficient is $k_{cooler} = 20$, and the corresponding pressure drop is evaluated using the liquid velocity v_l ;
- Heater: the loss coefficient is $k_{heater} = 40$, and the corresponding pressure drop is evaluated using the vapor velocity v_v .

The summation in the pressure loss equation can be explicitly expanded by accounting for the contributions of each section of the loop, applying the appropriate coefficients to the liquid and vapor regions:

$$(\rho_l - \rho_v)gh = \underbrace{\left(f_l \frac{L_l}{D} + k_{cooler} \right) \frac{1}{2} \rho_l v_l^2}_{\text{Losses in the liquid side}} + \underbrace{\left(f_v \frac{L_v}{D} + k_{heater} \right) \frac{1}{2} \rho_v v_v^2}_{\text{Losses in the vapor side}} \quad (9)$$

This formulation makes explicit the separation between liquid and vapor contributions, ensuring that each term is evaluated with the corresponding thermophysical properties and flow velocities.

²The contribution of body forces is determined by evaluating the scalar product $\int \rho g \hat{\mathbf{k}} \cdot d\mathbf{l} = \int \rho g \cos(\theta) dl$. In the horizontal segments of the circuit, the displacement vector $d\mathbf{l}$ is orthogonal to the gravity vector ($\theta = 90^\circ$). Consequently, as $\cos(90^\circ) = 0$, the integral over these sections vanishes and does not contribute to the driving head.

3 Determination of the Loop Height

Having established the governing equations and the formulation of pressure losses, the next step consists in determining the required elevation difference h between the hot and cold legs. This parameter represents the driving head of the natural circulation loop and must be sufficient to balance the total pressure losses in the system.

Since the geometry is assumed to be symmetric, the same height h is considered for both the hot and cold sections. This calculation provides a key design parameter for ensuring that the prescribed thermal power can be removed under steady-state natural circulation conditions.

3.1 Pipe Definition

The solution of the problem requires the assumption of a pipe diameter. Since the system under consideration is representative of an industrial application, this choice cannot be arbitrary, but must comply with established technical standards. In particular, pipe dimensions, wall thicknesses, and weights are defined by the international standard ANSI B36.10, which applies to both welded and seamless steel pipes.

As a preliminary assumption, a Nominal Pipe Size (NPS) of 10" has been selected. According to the standard, this corresponds to an outer diameter of 10.750 in and a wall thickness of 0.365 in.[2] While this initial choice will be further assessed and justified through subsequent numerical analysis, it provides a reasonable starting point for the calculations.

To ensure consistency with the International System of Units (SI), these quantities are converted into meters, yielding the following values:

- $d_{\text{out}} = 0.273 \text{ m}$
- $t = 9.27 \text{ mm}$
- $d_{\text{int}} = d_{\text{out}} - 2t = 0.254 \text{ m}$

3.2 Fluid Properties

Before proceeding with the solution, the fundamental thermophysical properties of the fluid under saturation conditions at a pressure of 70 bar were defined. These properties were obtained using the XSteam³ implementation in MATLAB, which provides reliable evaluations of water and steam properties based on established correlations.[3]

- Saturation temperature: $T_{\text{sat}} = 285.83 \text{ }^\circ\text{C}$
- Liquid density: $\rho_l = 739.72 \text{ kg/m}^3$
- Vapor density: $\rho_v = 36.52 \text{ kg/m}^3$
- Liquid specific enthalpy: $h_l = 1.2674 \times 10^3 \text{ kJ/kg}$
- Vapor specific enthalpy: $h_v = 2.7726 \times 10^3 \text{ kJ/kg}$
- Liquid dynamic viscosity: $\mu_l = 9.1249 \times 10^{-5} \text{ Pa} \cdot \text{s}$
- Vapor dynamic viscosity: $\mu_v = 1.8960 \times 10^{-5} \text{ Pa} \cdot \text{s}$

3.3 Friction Factor Calculation

The last parameter required for the solution of the Darcy–Weisbach equation (Eq. 6) is the friction factor f . This dimensionless quantity represents the resistance to fluid flow due to wall shear stress within the pipe and depends primarily on the Reynolds number (Re) and the relative roughness of the pipe (ε/D).

In the present analysis, the pipe is assumed to be hydraulically smooth, i.e., with negligible roughness ($\varepsilon = 0$). This additional simplification allows the friction factor to be evaluated using well-established semi-empirical correlations, depending on the flow regime:

³X Steam for Matlab is a implementation of the IAPWS IF97 standard formulation. It provides accurate data for water and steam and mixtures of water and steam properties from 0 to 1000 bar and from 0 to 2000 $^\circ\text{C}$.

$$f = \begin{cases} \frac{64}{Re} & \text{if } Re < 3000 \quad (\text{laminar flow}) \\ 0.316 Re^{-0.25} & \text{if } Re > 3000 \quad (\text{turbulent flow}) \end{cases} \quad (10)$$

Based on the available data and the assumptions made so far, the mass flow rate can be determined using the following relation:

$$\dot{m} = \frac{P_{th}}{h_v - h_l} = 23.12 \text{ kg/s}$$

Using the definition of mass flow rate, the velocities of both the liquid and vapor phases can be readily obtained as follows:

$$v_l = \frac{\dot{m}}{\rho_l A} = 0.61 \text{ m/s}$$

$$v_v = \frac{\dot{m}}{\rho_v A} = 12.44 \text{ m/s}$$

Finally, the Reynolds number can be evaluated using its classical definition:

$$Re_l = \frac{\rho_l v_l D}{\mu_l} = 1.27 \times 10^6$$

$$Re_v = \frac{\rho_v v_v D}{\mu_v} = 6.10 \times 10^6$$

Since both phases are characterized by Reynolds numbers well within the turbulent flow regime, the following friction factor values are obtained:

$$f_l = 0.0094 \quad (\text{cold leg})$$

$$f_v = 0.0064 \quad (\text{hot leg})$$

This preliminary analysis is thus concluded with the determination of the loop height:

$$h = \frac{(f_l \frac{L}{D} + k_{cooler}) \rho_l v_l^2 + (f_v \frac{L}{D} + k_{heater}) \rho_v v_v^2}{2g(\rho_l - \rho_v)} = 16.91 \text{ m}$$

4 Loop Height Calculation with Free Parametric Pipe Section Length

Secondly, the Loop Height calculation is repeated under the assumption that $h = L$, where the pipe section length L is treated as a free parameter. This introduces an additional unknown into the problem formulation.

As a consequence, the solution cannot be obtained analytically and must be determined through a numerical iterative procedure. The implementation of such a method requires the definition of a convergence tolerance, which has been set to 0.1 mm in order to ensure high accuracy, as well as an initial guess for h .

A reasonable initial estimate is $h = 16.91 \text{ m}$, corresponding to the value obtained in the previous section.

```

1 while err > tol && iter < max_iter
2     iter = iter + 1;
3
4     % 1. Distributed pressure losses (Pa) using current loop length L_iter
5     dp_fric_liquid = f_l * (L_iter / D) * 0.5 * rho_l * v_l^2;
6     dp_fric_vapor = f_v * (L_iter / D) * 0.5 * rho_v * v_v^2;
7
8     % 2. Total loop pressure losses
9     dp_loss_total = dp_fric_liquid + dp_fric_vapor + dp_conc_l + dp_conc_v;
```

```

10
11 % 3. Recalculation of the required loop height h
12 h_new = dp_loss_total / dp_buoyancy_spec;
13
14 % 4. Convergence error (absolute difference)
15 err = abs(h_new - L_iter)/abs(h_new);
16
17 % 5. Update of loop length for next iteration
18 L_iter = h_new;
19 end

```

As shown, the iterative solution was obtained through the implementation of a **while** loop. The final result yields a loop height of $h = 16.98$ m.

Furthermore, convergence is achieved after only two iterations, indicating that the initial estimate was already highly accurate.

5 Loop Height Calculation with Rough Pipe

Finally, the calculation of the loop height is repeated by removing the assumption of a hydraulically smooth pipe and considering the pipe length L as a free parameter. In this case, a finite absolute roughness $\varepsilon = 5.0 \times 10^{-5}$ m is introduced.

This modification directly affects the evaluation of the friction factor f , for which the correlations previously adopted (Eq. 10) are no longer applicable. For rough pipes, the friction factor depends on both the Reynolds number and the relative roughness, and must be determined using implicit or semi-empirical correlations such as:

- Colebrook equation:

$$\frac{1}{\sqrt{f}} = -2 \log_{10} \left(\frac{\varepsilon}{3.71D} + \frac{2.51}{Re\sqrt{f}} \right) \quad (11)$$

- Haaland equation:

$$\frac{1}{\sqrt{f}} = -1.8 \log_{10} \left[\left(\frac{\varepsilon}{3.7D} \right)^{1.11} + \frac{6.9}{Re} \right] \quad (12)$$

Alternatively, the friction factor can be estimated using the Moody diagram.

In the present analysis the friction factor is evaluated using the Haaland correlation (Eq 12), which provides an explicit and computationally efficient approximation differently from the Colebrook equation. The following values of the friction factor are obtained for the liquid and vapor phases, respectively:

$$f_l = 0.0144 \quad f_v = 0.0139$$

As expected, these values are higher compared to the smooth pipe assumption, due to the additional frictional effects introduced by the pipe roughness.

With the updated input parameters, the same iterative procedure described in the previous section is repeated. The resulting value of the loop height is $h = 17.20$ m.

The increase in the required height compared to the smooth pipe case is physically consistent. The presence of pipe roughness leads to higher friction factors, which in turn increase the total pressure losses along the loop. As a consequence, a larger buoyancy driving force is needed to sustain the same mass flow rate and ensure the removal of the prescribed thermal power. This is achieved by increasing the elevation difference h , which directly amplifies the hydrostatic pressure imbalance driving the natural circulation.

6 A Posteriori Assessment of Pipe Diameter

Before concluding, it is appropriate to assess the most suitable pipe diameter. To this end, six Nominal Pipe Sizes from the ANSI/ASME B36.10M standard (4", 5", 6", 8", 10", and 12") were considered, and the loop height required to sustain natural circulation was recalculated for each case, following the procedure described in Section 3.

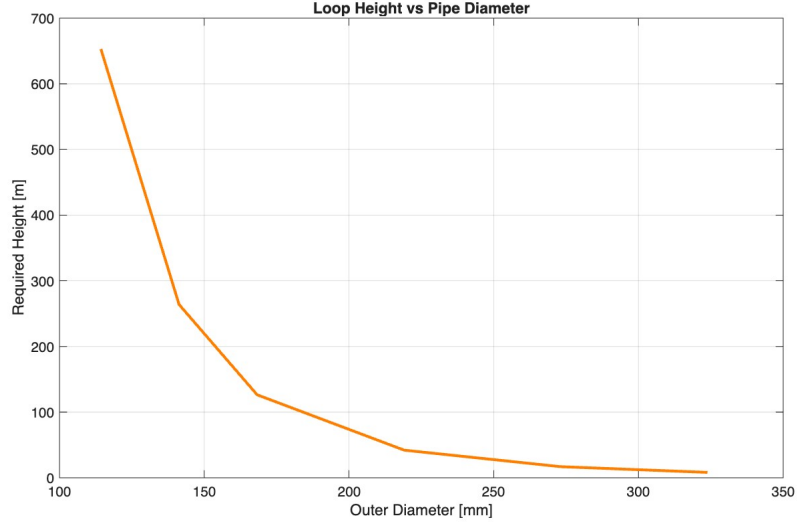


Figure 2: Variation of the required loop height h as a function of the pipe outer diameter for different nominal pipe sizes

NPS	Loop Height h [m]
4"	652.97
5"	263.87
6"	126.35
8"	42.07
10"	16.91
12"	8.21

Table 1: Loop height associated to NPS

As expected, varying the pipe diameter leads to different values of the loop height h . The corresponding trend is illustrated in Figure 3.

To improve the clarity and readability of the graph, the exact values of the loop height required for the considered Nominal Pipe Sizes (NPS) are reported in Table 1, based on the same assumptions outlined in Section 3.

It is immediately evident that, for pipe diameters smaller than the one adopted in the previous calculations, the required loop heights become impractically large, thus making the commercial deployment of the designed system nearly unfeasible.

Even the NPS of 10", although used as a reference in the previous analysis, results in a loop height that may still be considered relatively high and not suitable for all applications, unlike larger pipe diameters, which lead to more practical design solutions.

Conclusions

This study provides a physically coherent and analytically based model for designing a natural circulation loop operating at saturation conditions. Beginning with the Navier–Stokes equations, the analysis was simplified into a balance between buoyancy and hydraulic losses within the system.

An initial evaluation was carried out using a fixed geometric configuration, making it possible to estimate the loop height necessary to sustain the required thermal power. The model was then extended through a more flexible formulation in which the pipe section length was no longer fixed, but treated as a variable parameter. Because of this added unknown, a numerical iterative approach was adopted. The solution converged rapidly, supporting both the validity of the methodology and the reliability of the assumptions introduced in the model.

The influence of pipe roughness was also examined. The results showed that increasing surface roughness leads to higher frictional losses and, consequently, to a greater loop height requirement. This aspect underlines the importance of properly evaluating hydraulic resistance during the design phase of passive systems, where the available driving forces are naturally limited.

A further parametric investigation focused on the role of pipe diameter, revealing its strong influence on the overall feasibility of the system. Reduced diameters were associated with excessively large elevation requirements, making industrial manufacturing difficult. On the other hand, larger pipe diameters considerably decrease the necessary height difference, offering solutions that are more practical and adaptable from an engineering perspective. The balance between hydraulic efficiency and geometric limitations therefore emerges as a central design consideration for natural circulation loops.

References

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